

Sprint 6 — Analyst & User Guide

****N100 Financial Intelligence Platform****

API, Clustering & Quality Assurance — Days 36–45

1. Project Architecture

Sprint 6 builds two major capabilities on top of the N100 Financial Intelligence Platform:

1. ****KMeans Clustering Pipeline**** — Groups 92 Nifty-100 companies into 5 financially meaningful clusters using 5 key features.
2. ****REST API Layer**** — FastAPI-based read-only API serving company data, screener results, peer comparisons, portfolio analytics, and document access.

Directory Structure

```
Sprint-06-API-Clustering-QA/ ■■■■ src/ ■ ■■■■ analytics/ ■ ■ ■■■■ clustering.py # Day 36 – KMeans
clustering pipeline ■ ■ ■■■■ day37.py # Day 37 – Profiling, correlation, outliers, portfolio stats ■
■■■■ api/ ■ ■■■■ main.py # Day 38 – FastAPI app, CORS, logging, DB dependency ■ ■■■■ client.py # Day 42
– httpx-based API client for Streamlit ■ ■■■■ routers/ ■ ■■■■ health.py # /api/v1/health ■ ■■■■
companies.py # /api/v1/companies/* (7 endpoints) ■ ■■■■ screener.py # /api/v1/screener/* ■ ■■■■
sectors.py # /api/v1/sectors/* ■ ■■■■ peers.py # /api/v1/peers/* ■ ■■■■ valuation.py #
/api/v1/market-cap ■ ■■■■ portfolio.py # /api/v1/portfolio/* ■ ■■■■ documents.py # /api/v1/documents/*
■■■■ tests/ ■ ■■■■ test_clustering.py # 44 clustering tests ■ ■■■■ test_cluster_profiling.py # 15
profiling tests ■ ■■■■ test_portfolio_stats.py # 11 portfolio stats tests ■ ■■■■
test_outlier_detection.py # 13 outlier tests ■ ■■■■ test_data_quality.py # 53 data quality tests ■ ■■■■
api/ ■ ■■■■ test_health.py # 22 health/scaffold tests ■ ■■■■ test_companies.py # 50 company endpoint
tests ■ ■■■■ test_day40.py # 44 remaining endpoint tests ■ ■■■■ test_day42.py # 84 comprehensive
integration tests ■ ■■■■ test_day43_performance.py # 19 performance benchmark tests ■■■■ output/ ■ ■■■■
cluster_labels.csv # 92 company cluster assignments ■ ■■■■ cluster_profile.csv # 5 cluster profile
summaries ■ ■■■■ portfolio_stats.csv # 10 KPI portfolio statistics ■ ■■■■ outlier_report.csv # 912
sector outlier records ■■■■ reports/ ■ ■■■■ correlation_heatmap.png # 10-KPI Pearson correlation matrix
■ ■■■■ elbow_plot.png # KMeans inertia vs k curve ■ ■■■■ sprint6_test_report.html # Generated test
report ■ ■■■■ day43_performance_report.html # Performance benchmark report ■■■■ scripts/ ■■■■
generate_report.py # HTML test report generator ■■■■ generate_perf_report.py # HTML performance report
generator
```

2. Data Sources

Primary Database

- ****File:**** `Sprint-01-Data-Foundation/nifty100.db` (SQLite, 1.97 MB)
- ****Access:**** Read-only mode (`file:?mode=ro`)
- ****13 Tables:****

Table	Rows	Description
`companies`	92	Company master data (name, logo, ROE, ROCE, etc.)

Table	Rows	Description
`sectors`	92	Sector classifications, market cap categories
`financial_ratios`	1,164	Yearly KPI ratios (12+ years × 92 companies)
`profitandloss`	1,164	Annual P&L statements
`balancesheet`	1,058	Annual balance sheets
`cashflow`	1,063	Annual cash flow statements
`market_cap`	552	Market cap and valuation metrics
`peer_groups`	56	Peer group membership
`peer_percentiles`	560	Percentile-ranked peer comparisons
`documents`	1,456	Annual report links
`analysis`	16	Screeener/computed analysis
`prosandcons`	14	Pros/cons text (4 companies only)
`stock_prices`	5,520	Historical stock prices

Secondary Data Sources

- **Sprint-03:** `output/screener\_output.xlsx` — 6 screener presets (Quality Compounder, Value Pick, Growth Accelerator, Dividend Champion, Debt-Free Blue Chip, Turnaround Watch)
- **Sprint-05:** `reports/tearsheets/` — 91 pre-generated PDF tearsheets
- **Sprint-05:** `output/cashflow\_intelligence.xlsx` — Cash flow intelligence metrics (used by clustering)

3. Clustering Methodology

Features Used (5)

1. `return\_on\_equity\_pct` — Return on equity percentage
2. `debt\_to\_equity` — Debt-to-equity ratio
3. `revenue\_cagr\_5yr` — 5-year revenue CAGR
4. `fcf\_cagr\_5yr` — 5-year free cash flow CAGR
5. `operating\_profit\_margin\_pct` — Operating profit margin

Pipeline

1. **Data Loading:** Financial ratios (year=2024) + sector data from SQLite + cashflow intelligence from Excel
2. **Imputation:** Sector-median imputation for missing values (logged in-memory for validation)
3. **Scaling:** StandardScaler (zero mean, unit variance)
4. **Optimal k:** Elbow method (k=2..10), k=5 selected near elbow

5. **Clustering:** KMeans(n_clusters=5, random_state=42, n_init=10, max_iter=300)
6. **Naming:** Profile-based cluster naming using median feature values

Cluster Archetypes (5)

Cluster	Name	Count	Key Characteristics
0	Capital Efficient	60	Low D/E, moderate ROE, stable margins
1	Premium Margin Leaders	14	High OPM, strong ROE, moderate growth
2	Defense PSU — Extreme ROE Outlier	2	Extreme ROE (>500%), defense PSU
3	Financial Sector — High Leverage Growth	15	High D/E (leverage), strong growth, financial sector dominated
4	High FCF Growth Outlier	1	Exceptional FCF growth, high ROE

Determinism

- `random_state=42` ensures reproducibility
- Verified: repeated runs produce identical cluster assignments
- Cluster assignments are stable across runs

4. API Architecture

FastAPI Application

- **Title:** N100 Financial Intelligence Platform
- **Version:** 0.1.0
- **Base URL:** `http://127.0.0.1:8000`
- **Swagger UI:** `http://127.0.0.1:8000/docs`
- **OpenAPI JSON:** `http://127.0.0.1:8000/openapi.json`

Middleware

- **CORS:** Allow all origins (development mode)
- **Request Logging:** Logs method, path, status code, and response time for every request

Database Dependency

- Read-only SQLite connection per request
- Connection closed after request completes

- `sqlite3.Row` factory for dict-like access

5. API Endpoints (18 Total)

Health

Method	Path	Description
GET	`/api/v1/health`	Service status, uptime, version, DB row counts

Companies (7 endpoints)

Method	Path	Description
GET	`/api/v1/companies/`	List all 92 companies (filters: sector, market_cap_category, search)
GET	`/api/v1/companies/{ticker}`	Full company profile with latest ratios
GET	`/api/v1/companies/{ticker}/pl`	P&L history (filters: from_year, to_year)
GET	`/api/v1/companies/{ticker}/bs`	Balance sheet history (filters: from_year, to_year)
GET	`/api/v1/companies/{ticker}/cashflow`	Cashflow history (filters: from_year, to_year)
GET	`/api/v1/companies/{ticker}/ratios`	Financial KPI ratios (filter: year)
GET	`/api/v1/companies/{ticker}/tearsheet`	Pre-generated tearsheet PDF

Screener

Method	Path	Description
GET	`/api/v1/screener/`	List 6 screener presets with company counts
GET	`/api/v1/screener/{preset}`	Companies matching a preset

Sectors

Method	Path	Description
GET	`/api/v1/sectors/`	List 10 broad sectors with company counts

Method	Path	Description
GET	<code>`/api/v1/sectors/{sector}/companies`</code>	Companies in a sector (filter: market_cap_category)

Peers

Method	Path	Description
GET	<code>`/api/v1/peers/{ticker}`</code>	Peer group membership
GET	<code>`/api/v1/peers/{ticker}/compare`</code>	Percentile-ranked peer comparison (filter: year)

Market Cap

Method	Path	Description
GET	<code>`/api/v1/market-cap/`</code>	Market cap data (filters: year, sector, market_cap_category)

Portfolio

Method	Path	Description
GET	<code>`/api/v1/portfolio/stats`</code>	Portfolio statistics for 10 KPIs
GET	<code>`/api/v1/portfolio/clusters`</code>	Cluster assignments for 92 companies

Documents

Method	Path	Description
GET	<code>`/api/v1/documents/{ticker}`</code>	Annual report documents (filter: year)

6. Screener Presets

Preset	Description	Data Source
Quality Compounder	High ROE, consistent growth	Sprint-03 screener_output.xlsx
Value Pick	Low PE, high dividend yield	Sprint-03 screener_output.xlsx
Growth Accelerator	High revenue CAGR	Sprint-03 screener_output.xlsx

Preset	Description	Data Source
Dividend Champion	High dividend payout	Sprint-03 screener_output.xlsx
Debt-Free Blue Chip	Zero or low D/E	Sprint-03 screener_output.xlsx
Turnaround Watch	Improving financials	Sprint-03 screener_output.xlsx

7. Portfolio Statistics

10 key performance indicators with distribution statistics:

Metric	Description
net_profit_margin_pct	Net profit as % of revenue
operating_profit_margin_pct	Operating profit as % of revenue
return_on_equity_pct	Return on shareholders' equity
debt_to_equity	Total debt / shareholders' equity
return_on_capital_employed_pct	ROCE percentage
asset_turnover	Revenue / total assets
revenue_cagr_5yr	5-year revenue compound annual growth rate
fcf_conversion_pct	Free cash flow / operating profit
cfo_quality_score	Cash from operations / net profit
capex_intensity_pct	Capital expenditure / revenue

For each metric: P10, P25, P50 (median), P75, P90, Mean, Std, n_valid, n_missing.

8. How to Start the API

```
cd Sprint-06-API-Clustering-QA # Start the FastAPI server uvicorn src.api.main:app --host 127.0.0.1
--port 8000 --reload # Access Swagger UI open http://127.0.0.1:8000/docs # Access OpenAPI JSON curl
http://127.0.0.1:8000/openapi.json
```

9. How to Use the API Client

The API client (`src/api/client.py`) provides typed helper functions for Streamlit integration:

```
from src.api.client import ( api_health, api_get_companies, api_get_company, api_get_pl, api_get_bs,
api_get_cashflow, api_get_ratios, api_get_tearsheet, api_get_screener_presets,
api_get_screener_results, api_get_sectors, api_get_sector_companies, api_get_peers,
api_get_peer_compare, api_get_market_cap, api_get_portfolio_stats, api_get_portfolio_clusters,
api_get_documents, ) # Example usage companies = api_get_companies(sector="Financials") tcs =
api_get_company("TCS") pl = api_get_pl("TCS", from_year="2020") tearsheet_pdf =
api_get_tearsheet("TCS") # Returns bytes or None
```

****Requirements:**** FastAPI server must be running on `http://127.0.0.1:8000`.

10. How to Run Tests

```
cd Sprint-06-API-Clustering-QA # Run all tests
pytest tests/ -v # Run specific test modules
pytest tests/test_clustering.py -v # Clustering tests
pytest tests/api/test_day42.py -v # Integration tests
pytest tests/api/test_day43_performance.py -v -s # Performance benchmarks
# Generate HTML test report
python scripts/generate_report.py
```

11. Performance Benchmarks

Local benchmark results (20 requests per endpoint, 3 warmup):

Rank	Endpoint	Avg Latency	Median	P95
1	screeener	15.1ms	14.6ms	16.6ms
2	market_cap	7.0ms	7.0ms	7.4ms
3	ratios	5.0ms	3.4ms	34.8ms
4	portfolio_clusters	4.4ms	4.3ms	6.0ms
5	peers_compare	3.0ms	2.8ms	5.7ms
6-13	all others	2.0–2.9ms	—	—

****Note:**** These are LOCAL BENCHMARK MEASUREMENTS, not production-grade performance claims.

12. Known Data-Quality Exceptions

Exception	Count	Explanation
JIOFIN no tearsheet	1	Listed in 2023, < 3 years history, legitimately skipped by Sprint-05
Companies with pros/cons	4/92	Only 4 companies have pros/cons data in the database
Companies without peer groups	36/92	56 companies have peer group membership
Missing from documents	1	One company has no document records
Extreme financial ratios	~10	ROE > 500% (BEL, HAL, INDIGO), ROCE > 500% (5 companies), NPM > 200% (BAJAJHLDNG) — legitimate outliers, not data errors
Database tables	13	All 13 tables are legitimate and used

13. Database Indexes

All queries are covered by existing indexes:

Index	Table	Columns	Purpose
`idx_pnl_company`	profitandloss	company_id	P&L queries by company
`idx_bs_company`	balancesheet	company_id	Balance sheet queries
`idx_cf_company`	cashflow	company_id	Cashflow queries
`idx_fr_company`	financial_ratios	company_id	Ratio queries
`idx_mc_company`	market_cap	company_id	Market cap queries
`idx_prices_company_date`	stock_prices	company_id, date	Stock price queries
`idx_sectors_broad`	sectors	broad_sector	Sector listing
`idx_pp_company`	peer_percentiles	company_id	Peer lookups
`idx_pp_group`	peer_percentiles	peer_group_name	Peer compare

****No new indexes required.**** With 92 companies, all queries complete in under 16ms.

14. Final Acceptance Status — Sprint 6

****Status: CONDITIONAL PASS****

All 20 acceptance gates pass. Three upstream data-coverage exceptions are acknowledged but are ****not**** Sprint 6 defects — they originate from source data pipelines (Sprint-01, Sprint-05) and do not block API functionality or clustering correctness.

Upstream Data Exceptions (Not Sprint 6 Defects)

#	Exception	Impact	Origin
1	JIOFIN tearsheet missing	`/companies/JIOFIN/tearsheet` returns 404	Sprint-05 — company listed 2023, <3yr history
2	4/92 companies have pros/cons	Peer compare pros/cons sparse	Sprint-01 — only 4 companies populated
3	36/92 companies lack peer groups	Peer endpoints return 404 for 36 tickers	Sprint-01 — only 56 companies in peer_groups

Test Suite Summary

- ****Collected:**** 355
- ****Passed:**** 352
- ****Skipped:**** 3 (data-coverage gaps, upstream)
- ****Failed:**** 0